ADC In Vivo Biotransformation & PK Report

2026-04-25

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1 Study Information

Field	Value
Study ID	ADC-PK-001 [SYNTHETIC DEMO DATA – not from a real study]
Compound	ADC-001
Species	Cynomolgus monkey
Dose	3.6 mg/kg IV
Nominal DAR	3.5
Concentration Unit	ug/mL
Timepoints	11
Time Range	0.083 – 672 h
Report Date	2026-04-25
Software	R 4.4.2 / PKNCA 0.12.1

2 PK Concentration-Time Profiles

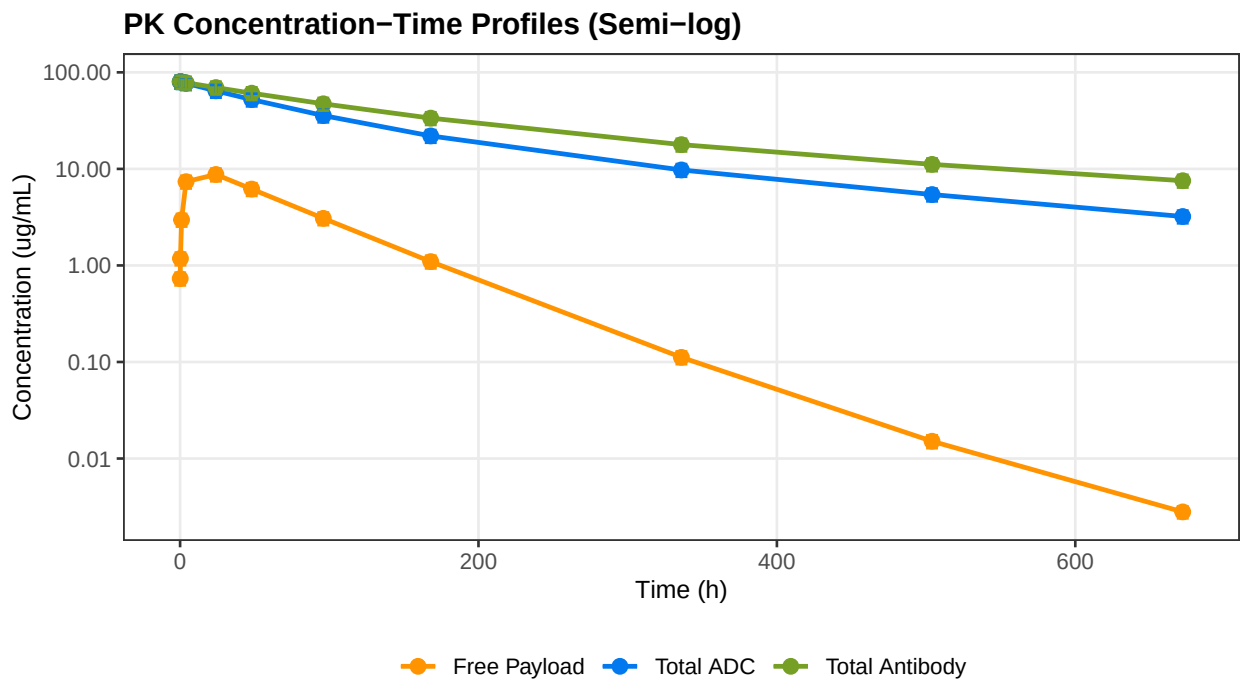

Figure 1: Multi-analyte PK concentration-time profiles on semi-log scale. Points represent mean concentrations; error bars show +/-SD where available.

3 DAR Trajectory

3.1 Average DAR Over Time

Average DAR was calculated as:

$$\text{avg DAR}(t) = \frac{\text{Total ADC}(t)}{\text{Total Antibody}(t)} \times \text{Nominal DAR}$$

where both analytes are measured in mass concentration (ug/mL). The ratio gives fractional conjugation; multiplying by nominal DAR converts to drug molecules per antibody molecule.

3.2 DAR Stability Summary

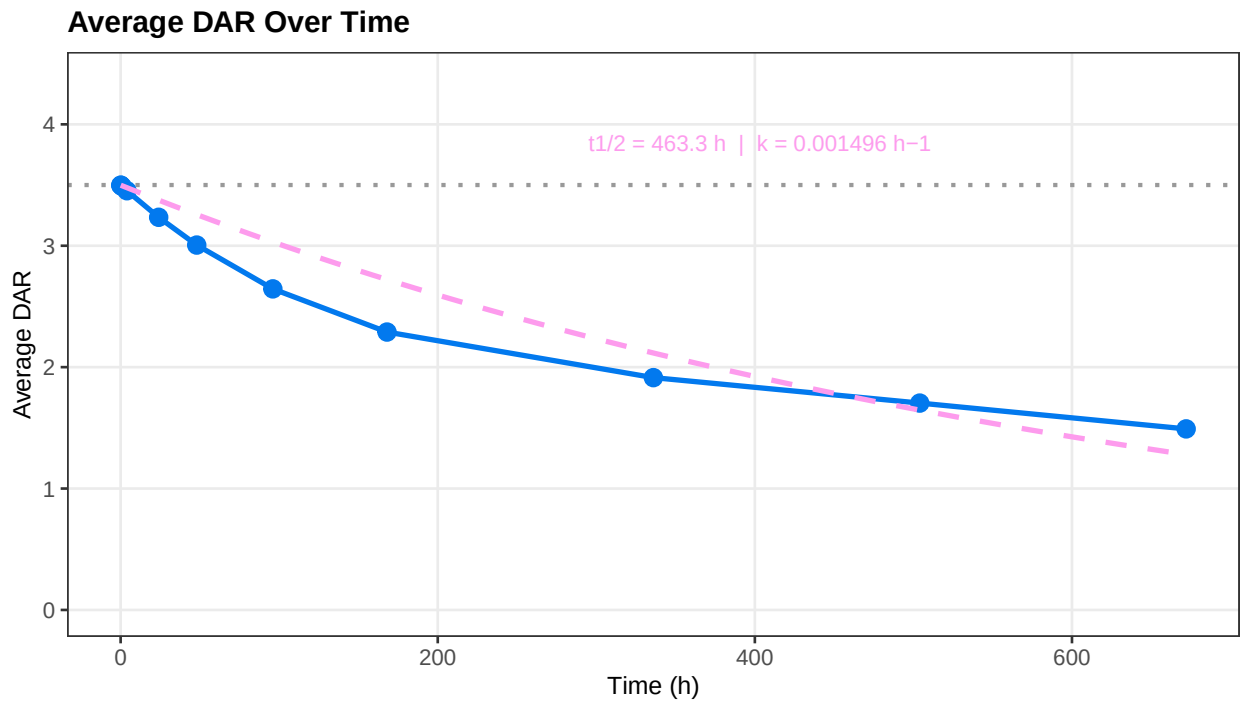

Figure 2: Average DAR over time. Dashed line: exponential decay fit. Dotted line: nominal DAR at dose.

Table 2: DAR stability metrics

Metric	Value
Nominal DAR	3.5
DAR at T-first	3.499
DAR at T-last	1.491
% Nominal DAR Retained	42.6
Decay Rate k (h-1)	0.001496
DAR Half-life (h)	463.3

3.3 DAR Trajectory Data

Table 3: Average DAR and fractional conjugation over time

Time (h)	Total ADC	Total Ab	Frac. Conjugation	Avg DAR	% Nominal DAR
0.083	79.938	79.961	1.000	3.499	99.971
0.250	79.814	79.882	0.999	3.497	99.915
1.000	79.260	79.530	0.997	3.488	99.661
4.000	77.089	78.139	0.987	3.453	98.656
24.000	64.319	69.604	0.924	3.234	92.407
48.000	52.261	60.856	0.859	3.006	85.876
96.000	35.691	47.233	0.756	2.645	75.564

Time (h)	Total ADC	Total Ab	Frac. Conjugation	Avg DAR	% Nominal DAR
168.000	21.957	33.571	0.654	2.289	65.405
336.000	9.752	17.836	0.547	1.914	54.676
504.000	5.424	11.141	0.487	1.704	48.685
672.000	3.214	7.543	0.426	1.491	42.609

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4 Free Payload Release Kinetics

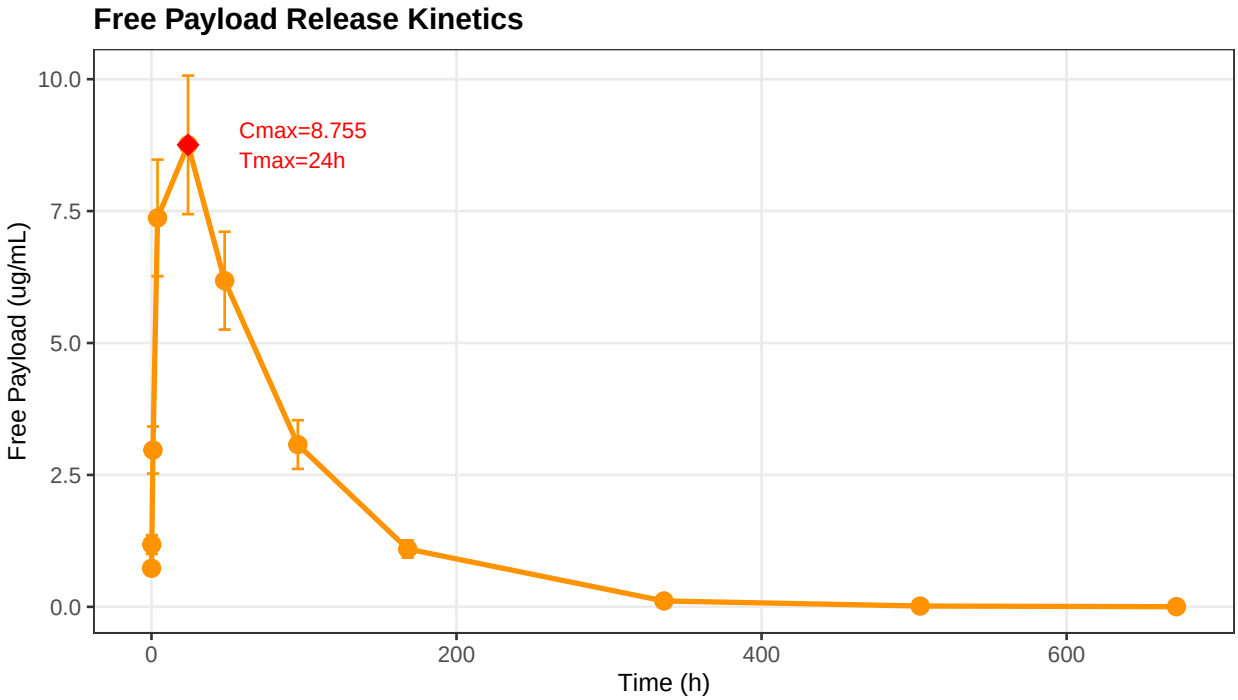

Figure 3: Free payload concentration-time profile. Cmax and Tmax annotated.

Table 4: Free payload PK highlights

Metric	Value
Cmax	8.7550
Tmax (h)	24.0000
Clast	0.0028
AUClast (trapezoidal)	843.6355

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5 PK/PD Overlay

The following dual-axis plot overlays the average DAR trajectory (left axis) with free payload concentration (right axis) to visualise the temporal relationship between ADC deconjugation and payload systemic exposure.

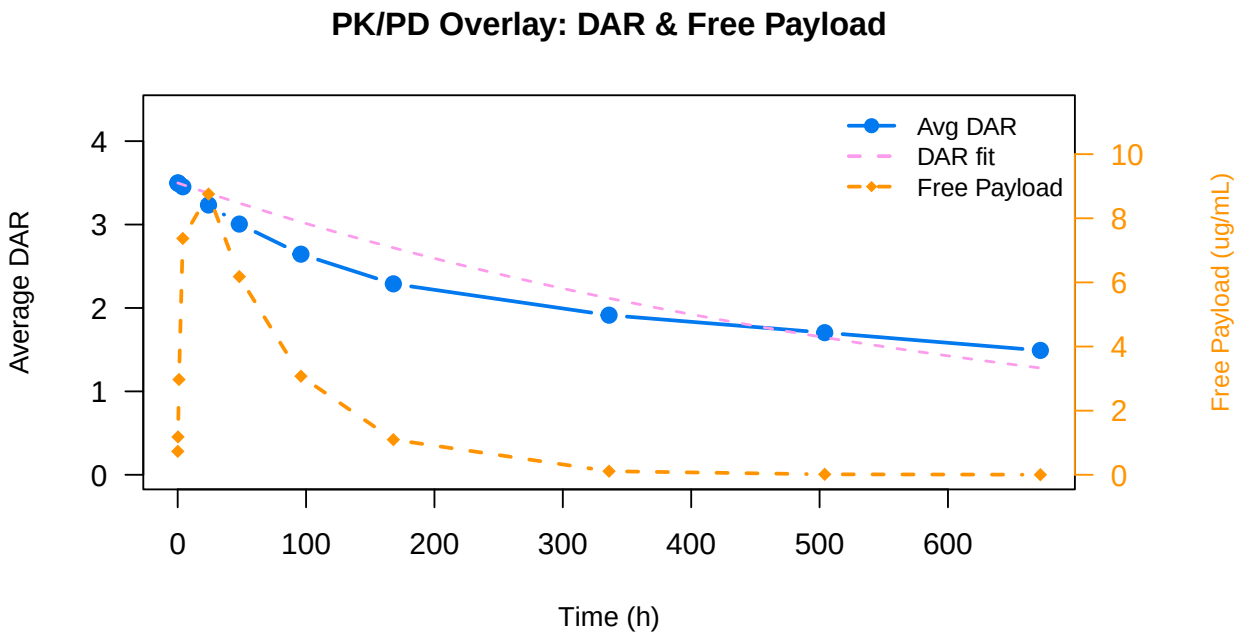

Figure 4: PK/PD overlay: average DAR trajectory (left axis, blue) and free payload concentration (right axis, orange). Dashed line: exponential DAR decay fit.

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6 NCA Parameters

Note on tmax: A tmax value of 0 indicates that Cmax occurs at the first measured timepoint within the NCA interval (T-first = first observed time, not nominal time zero). This is expected for IV bolus administration where peak concentrations are observed at the earliest sampling time.

6.1 Total ADC

Table 5: NCA parameters — Total ADC

Parameter	Value
auclast	11708.0587
cmax	79.9380
tmax	0.0000
tlast	671.9170

Parameter	Value
lambda.z	0.0033
r.squared	0.9989
adj.r.squared	0.9978
lambda.z.time.first	335.9170
lambda.z.time.last	671.9170
lambda.z.n.points	3.0000
clast.pred	3.1803
half.life	209.8258
span.ratio	1.6013

6.2 Total Antibody

Table 6: NCA parameters — Total Antibody

Parameter	Value
auclast	16929.7002
cmax	79.9610
tmax	0.0000
tlast	671.9170
lambda.z	0.0026
r.squared	0.9971
adj.r.squared	0.9942
lambda.z.time.first	335.9170
lambda.z.time.last	671.9170
lambda.z.n.points	3.0000
clast.pred	7.4424
half.life	270.6225
span.ratio	1.2416

6.3 Free Payload

Table 7: NCA parameters — Free Payload

Parameter	Value
auclast	789.3524
cmax	8.7550
tmax	23.9170
tlast	671.9170
lambda.z	0.0110
r.squared	0.9976
adj.r.squared	0.9951
lambda.z.time.first	335.9170
lambda.z.time.last	671.9170
lambda.z.n.points	3.0000

Parameter	Value
clast.pred	0.0027
half.life	63.1810
span.ratio	5.3181

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7 Methods

DAR Calculation: Average DAR at each timepoint was computed as the ratio of Total ADC to Total Antibody concentrations (both in ug/mL) multiplied by the nominal DAR at dose (3.5). This approach assumes both analytes are measured by mass-based assays (e.g., ELISA or LC-MS/MS) with equivalent antibody backbone molecular weight.

DAR Decay Fit: An exponential decay model DAR(t) = DAR0 x exp(-k x t) was fitted to the observed DAR trajectory using nonlinear least squares (R `nls()`). DAR half-life was derived as t1/2 = ln(2)/k.

Non-Compartmental Analysis: NCA was performed using the PKNCA package (v0.12.1) in R. The analysis interval spanned from the first observed timepoint to the last observed timepoint. AUClast was computed by the linear-log trapezoidal method. Terminal half-life was estimated from the log-linear terminal phase using automated lambda-z selection.

Payload Release Rate: Payload release rate (DeltaC/Deltat) was estimated as the finite difference between consecutive timepoints.

Software: R 4.4.2 | PKNCA 0.12.1 | ggplot2 | openxlsx

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